Rishikesh Galande

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SUMMARY

Cybersecurity graduate student with hands-on experience designing and operating secure networked systems, intrusion detection pipelines, and defensive DNS infrastructure. Strong foundation in TCP/IP, DNS, Linux networking, and packet analysis, with practical experience building IDS environments, tuning detection logic, and analyzing real-world traffic. Proven ability to automate security workflows, investigate network threats, and deploy resilient, privacy-focused network services using Snort, Splunk, Docker, and Python.

EDUCATION

Pennsylvania State University — **State College, PA**
Master of Science in Cybersecurity Analytics & Operations, GPA: 4.0/4.0 — *May 2026*

Ramrao Adik Institute of Technology — **Mumbai, Maharashtra**
Bachelor of Technology in Information Technology — *May 2024*

SKILLS

- **Networking & Infrastructure:** TCP/IP, DNS, Packet Analysis, Network Traffic Analysis, Firewalls, Routing & Switching, Linux Networking, IDS/IPS
- **Security Operations & Detection:** SOC Operations, Incident Response, Intrusion Detection, Threat Detection, Log Analysis, Alert Triage
- **Tools & Systems:** Snort, Splunk (SIEM), Linux, Windows, Docker, System Hardening, IDS Rule Development, Python Automation.

WORK EXPERIENCE

Pennsylvania State University — **State College, PA**
Cybersecurity Analytics Teaching Assistant (SOC & Network Security Labs) — *August 2025 - Present*

- Led hands-on cybersecurity labs in security operations and network forensics using Splunk (SIEM), Wireshark, and Python, guiding students through real-world traffic analysis and intrusion detection.
- Mentored 50+ students in automation, log parsing, and investigation workflows, improving incident detection speed by 30% and reducing recurring misconfigurations by 20%.
- Reviewed and graded 100+ labs while supporting fault isolation, root-cause analysis, and the development of clear lab documentation and troubleshooting guide.

PROJECTS

Network Intrusion Detection Lab | Pennsylvania State University
Docker, Snort, Tcpdump, Wireshark, Python, Linux

- Designed a SHA-256–based deterministic polymorphic Network Intrusion Detection System (NIDS) lab using Docker, Snort, Python, and Linux to generate unique, reproducible attack parameters per user.
- Simulated realistic network attacks and background traffic using hping3, tcpdump, packet analysis, TCP/UDP scanning, and probabilistic traffic modeling to support IDS detection and forensics.
- Automated IDS rule validation, security testing, incident analysis workflows, and containerized deployments, achieving 91% system quality, 94% test coverage, and fully reproducible environments.

Defensive DNS Infrastructure | Independent Project
DNS, Unbound, Raspberry Pi, Linux, DNSSEC

- Designed a privacy-first defensive DNS infrastructure using Pi-hole, Unbound, Cloudflared, DNSSEC, and Linux networking to provide network-wide ad, malware, and phishing protection.
- Built a recursive DNS resolution pipeline with Unbound querying root servers directly, enforcing strict resolution order and eliminating third-party resolver privacy leakage.
- Optimized and hardened DNS services on a Raspberry Pi Zero 2 W through cache tuning, memory management, NAT configuration, and encrypted DoH failover, achieving high uptime and low-latency resolution.

CERTIFICATIONS

- CompTIA Security+
- AWS Certified Cloud Practitioner
- Cisco Certified Network Associate (CCNA) – In Progress